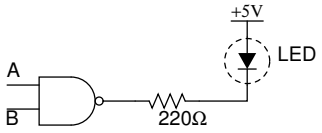
Digital Laboratory - II

Using LEDs and seven segment displays

Design a circuit to convert a 4 bit binary number to its BCD representation. Display the BCD number on a seven segment display. You can use *a single* 7447 driver, which is a BCD to seven segment decoder. Include ripple blanking. Read the data sheet carefully before wiring the circuit.

NOTE: A TTL gate can typically sink more current than it can source. So connect the LED so that it glows when the output is low. An example is shown below. Do not forget the current limiting resistor.



Procedure:

Binary to BCD Conversion:

Simple logic for this conversion is

1. If the Binary Number is 9 or less than 9, BCD representation is same as the Binary.
2. If the Binary Number is greater than 9, add 6 to the Binary number to get its BCD equivalent.

For Example: BCD equivalent of Binary Digit C is 1100 + 0110 = 10010 in BCD

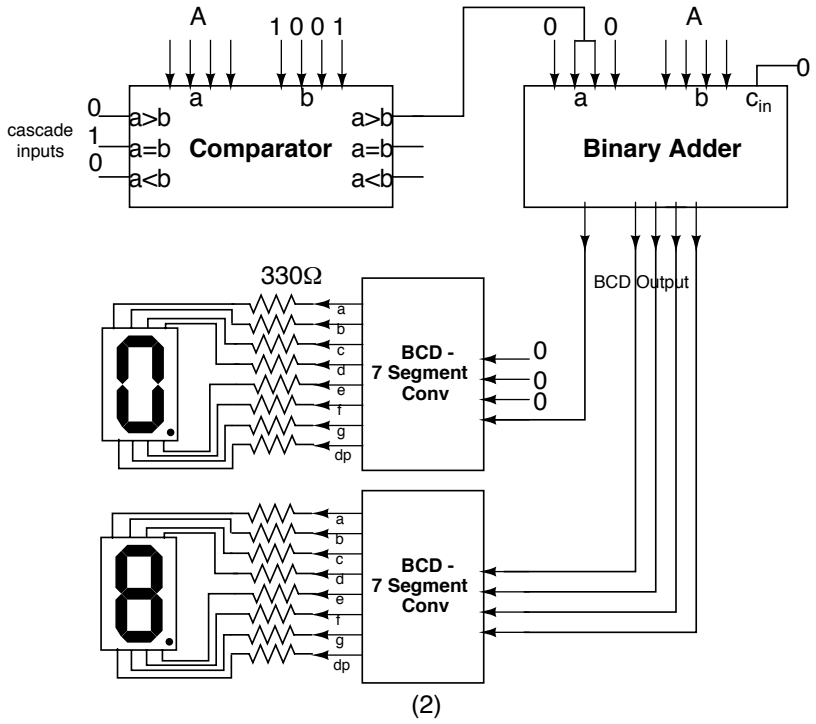


Figure 1: BCD to Seven Segment Display Circuit

Figure shows the circuit diagram to implement the above logic. The output BCD digits are then converted to 7-Segment code using IC 7483 for 7-Segment display. The 7-segment displays used in the lab are of common anode type i.e. anodes of all the LEDs are tied to V_{cc}. Typical operating point of each segment LED varies from 1.6V to 2V at 20mA forward current. The resistors from IC 7483 to the segments have to be chosen in such a way that the current drawn by the LEDs do not exceed 20mA.

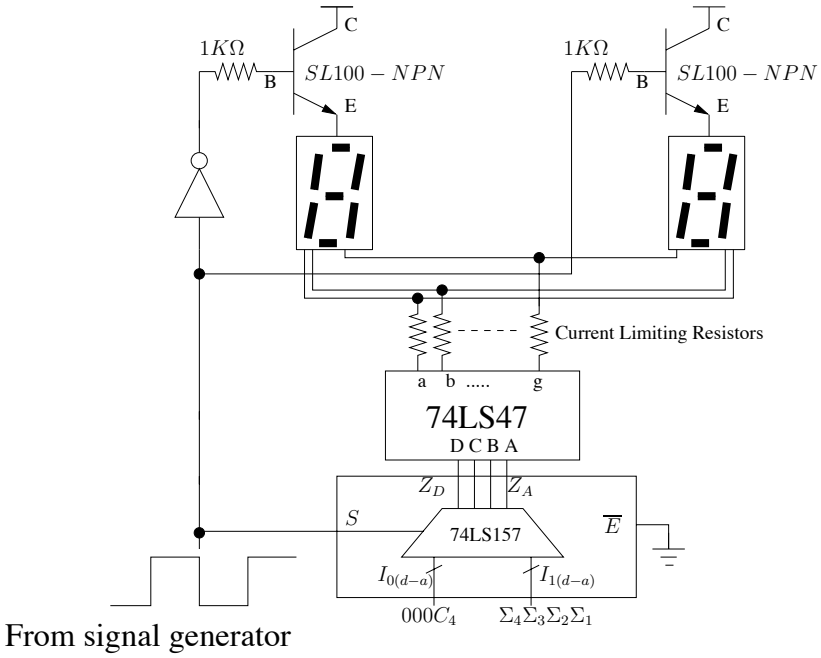


Figure 2: Controlling multiple seven segment displays with one driver - Comparator and adder circuit is identical to earlier case. Inverter (NOT gate) is from 74LS04